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Using K-maps to build binary adders



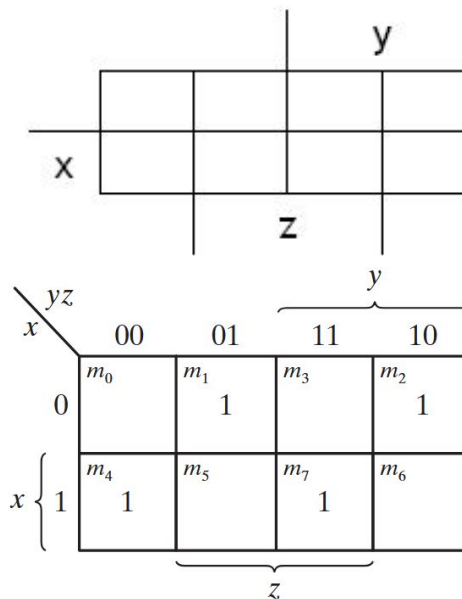
Half Adder

x	y	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Full Adder

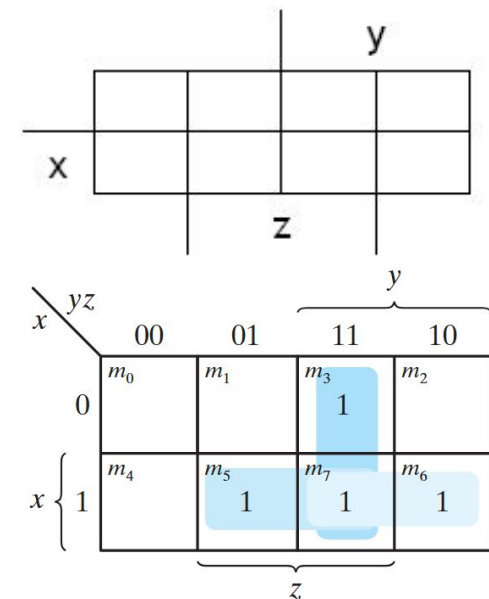
x	y	z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

K-map for S



$$(a) S = x'y'z + x'yz' + xy'z' + xyz$$

K-map for C



$$(b) C = xy + xz + yz$$

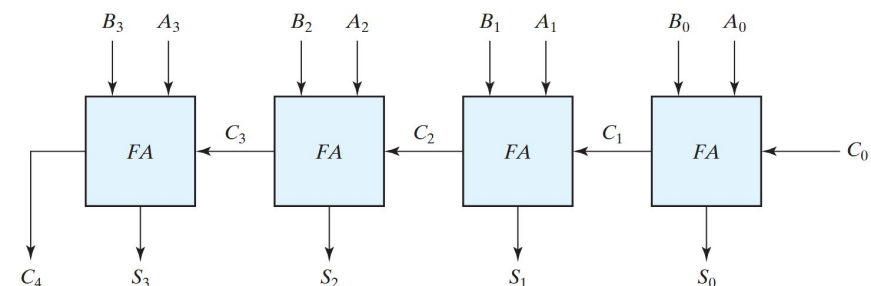


FIGURE 4.9
Four-bit adder

Four variable K-maps

- We can do four-variable Karnaugh maps too!
- A four-variable function $f(w,x,y,z)$ has sixteen possible minterms. They can be arranged so that adjacent minterms have common literals.
 - You can wrap around the sides *and* the top and bottom.
 - Again the minterms are almost, but not quite, in numeric order.

		y				
		$w'x'y'z'$	$w'x'y'z$	$w'x'y z$	$w'x'y z'$	
		$w'x y'z'$	$w'x y'z$	$w'x y z$	$w'x y z'$	
w		$w x y'z'$	$w x y'z$	$w x y z$	$w x y z'$	x
		$w x'y'z'$	$w x'y'z$	$w x'y z$	$w x'y z'$	
		z				

		y				
		m_0	m_1	m_3	m_2	
		m_4	m_5	m_7	m_6	
w		m_{12}	m_{13}	m_{15}	m_{14}	x
		m_8	m_9	m_{11}	m_{10}	
		z				

Four variable example

- Let's say we want to simplify $m_0 + m_2 + m_5 + m_8 + m_{10} + m_{13}$

		y				
		m_0	m_1	m_3	m_2	
		m_4	m_5	m_7	m_6	x
		m_{12}	m_{13}	m_{15}	m_{14}	
w		m_8	m_9	m_{11}	m_{10}	
		z				

		y				
		1	0	0	1	
		0	1	0	0	x
		0	1	0	0	
w		1	0	0	1	
		z				

- The following groups result in the minimal sum of products $x'z' + xy'z$.

		y				
		1	0	0	1	
		0	1	0	0	x
		0	1	0	0	
w		1	0	0	1	
		z				

Prime implicants

- Finding the best groups is even more difficult in larger K-maps.
- One good approach to deriving an MSP is to first find the largest possible groupings of minterms.
 - These groups correspond to **prime implicant** terms.
 - The final MSP will contain a subset of the prime implicants.
- Here is an example K-map with prime implicants marked.

					y
		1	1	0	0
		1	1	0	0
		0	1	1	0
w		0	0	1	1
					z

Essential prime implicants

- If any minterm belongs to only one group, then that group represents an **essential prime implicant**.
- Essential prime implicants *must* appear in the final MSP, which has to include all of the original minterms.

				y	
		1	1	0	0
		1	1	0	0
					x
w	0	1	1	0	
	0	0	1	1	
					z

- This example has two essential prime implicants.
 - The red group ($w'y$) is essential, since m_0 , m_1 and m_4 are not in any other group.
 - The green group ($wx'y$) is essential because of m_{10} .

Covering the remaining minterms



- Finally, pick as few other prime implicants as necessary to ensure that all of the original minterms are included.

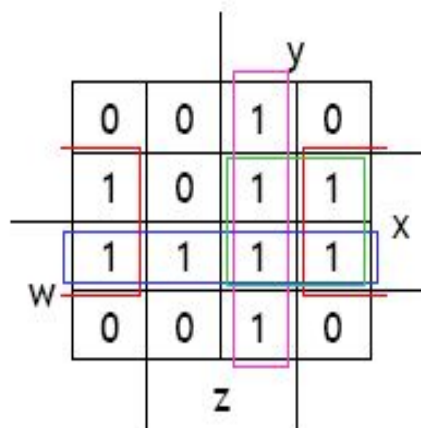
		y			
		1	1	0	0
		1	1	0	0
w	0	1	1	0	
	0	0	1	1	
		z		x	

- After choosing the red and green rectangles in our example, there are just two minterms remaining, m_{13} and m_{15} .
 - They are both included in the blue prime implicant, wxz .
 - The resulting MSP is $w'y' + wxz + wx'y$.
- The magenta and sky blue groups are not needed, since their minterms are already included by the other three prime implicants.

Example 2



- Simplify the following K-map.



- All prime implicants are circled.
- The essential prime implicants are xz' , wx and yz .
- The MSP is $xz' + wx + yz$. (Including the group xy would be redundant.)

Don't Care Conditions

- Unspecified outputs for certain inputs
- Used in Map to provide further simplification
- X is marked inside the square for don't care input
- Choose to include each don't care minterm as either 1 or 0

Don't care conditions

- There are times when we don't care what a function outputs—some input combinations might never occur, or some outputs may have no affect.
- We can express these situations with **don't care conditions**, denoted with **X** in truth table rows.
- An expression for this function has two parts.
 - One expression corresponds to outputs of 1.
 - Another describes the don't care conditions.

$$f(x,y,z) = m_3, d(x,y,z) = m_2 + m_4 + m_5$$

- Circuits *always* output 0 or 1; there is no value called "X". Instead, the Xs just indicate cases where both 0 or 1 would be acceptable outputs.

x	y	z	f(x,y,z)
0	0	0	0
0	0	1	0
0	1	0	X
0	1	1	1
1	0	0	X
1	0	1	X
1	1	0	0
1	1	1	0

Don't care simplifications

- In a K-map we can treat each don't care as 0 or 1. Different selections can produce different results.

			y	
	0	0	1	X
x	X	X	0	0
			z	

- In this example we can use the don't care conditions to our advantage.
 - It's best to treat the bottom two Xs as 0s. If either of them were 1, we'd end up with an extra, unnecessary term.
 - On the other hand, interpreting the top X as 1 results in a larger group containing m_3 .
- The resulting MSP is $x'y$.

Example 3

- Find a minimal sum of products for the following.

$$f(w,x,y,z) = \sum m(0,2,4,5,8,14,15), \quad d(w,x,y,z) = \sum m(7,10,13)$$

					y
	1	0	0	1	
	1	1	X	0	
	0	X	1	1	x
w	1	0	0	X	
					z

- All prime implicants are circled. We can treat Xs as 1s if we want, so the red group includes two Xs, and the light blue group includes one X.
- The *only* essential prime implicant is $x'z'$. The red group is not essential because the two minterms in it also appear in other groups.
- The MSP is $x'z' + wxy + w'xy'$. It turns out the red group is redundant; we can cover all of the minterms in the map without it.

POS Simplification

- Combine valid adjacent squares containing 0's
- Obtain simplified expression of the complement in SOP form
- Compliment further to get function in POS form

$$F(A,B,C,D) = \Sigma(0,1,2,5,8,9,10)$$

1	1	0	1
0	1	0	0
0	0	0	0
1	1	0	1

$$F' = AB + CD + BD'$$

By De Morgan's:

$$F = (A'+B')(C'+D')(B'+D)$$

Five variable K-map

$A = 0$

		DE		D	
		00	01	11	10
BC	00	0	1	3	2
	01	4	5	7	6
	11	12	13	15	14
	10	8	9	11	10

E

C

B

$A = 1$

		DE		D	
		00	01	11	10
BC	00	16	17	19	18
	01	20	21	23	22
	11	28	29	31	30
	10	24	25	27	26

E

C

B